

KANMANI BHARATHI J

Molecular Biology and Biotechnology Researcher

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PORTFOLIO

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Researcher ID : GOE-5646-2022

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GitHub : [kanmanibharathi](https://github.com/kanmanibharathi)

PROFILE SUMMERY

Dr. Kanmani Bharathi J is a Plant Molecular Biology and Biotechnology researcher specializing in medicinal plant biotechnology, tissue culture, micropropagation, phytochemistry, and natural-product research. His expertise spans *in vitro* culture optimization, callus induction and regeneration, aseptic techniques, metabolite profiling using GC-MS and LC-MS, transcriptomics, bioinformatics, molecular pharmacology, and bioactivity evaluation. He also has strong capabilities in experimental design, statistical analysis, and data visualization using R and Python, with experience integrating computational approaches into biological research. His interdisciplinary work extends to plastic biodegradation, microbial biotechnology, and the development of data-driven research and statistical analysis tools. He is committed to innovation-oriented research, reproducible experimental systems, scientific problem-solving, and translating biological discoveries into practical applications.

RESEARCH EXPERIENCE

Ph.D. Researcher [Annamalai University, 2021 - 2025]

- Investigated the antidiabetic properties of natural compounds from medicinal plants.
- Optimized extraction methods
- Conducted analytical analysis to identify potential drug candidates.
- Studied glucose uptake and insulin response mammalian cell lines.
- In silico approaches
- Micropropagation of medicinal plants.

Research Contributed [Annamalai University, 2023-2024]

- Worked on microbial degradation of plastics using bacteria.

MISCELLANEOUS EXPERIENCE

- Founder & Research Consultant [ZYNDORA, 2024 – 2025]
- Science Content Creator [INSIGHTS – 2021 – 2022]

TECHNICAL SKILLS

- **Molecular Techniques:** PCR, molecular markers, protein analysis (western blotting).
- **Biotechnology:** Micropropagation, somatic hybridization, and cell line studies.
- **Natural products:** Phytochemical analysis and extraction, metabolite profiling.
- **Instrumentation & Analysis:** LC-MS, GC-MS, spectroscopy.
- **Computational Techniques:** Molecular docking, geo statistics (Beginner), bioinformatics (Beginner).
- **Graphical Illustration:** Adobe creative tools, BioRender, Canva affinity.
- **Engineering & Electronics:** Automation tools, microcontrollers, custom research instrument design.
- **Software:** Microsoft tools, language (R and Python (Beginner)), Statistical tools (GraphPad Prism, Origin, SPSS, R studio and shiny).

- AI tools for research.

EDUCATION

Doctor of Philosophy (Ph.D.) in Plant Molecular Biology and Biotechnology,
Annamalai University, Chidambaram, Tamil Nadu, India.

Graduated, **Sep 2025**
OGPA: **9.25**

Master's in Plant Molecular Biology and Biotechnology,
Annamalai University, Chidambaram, Tamil Nadu, India.

Graduated, **May 2021**
OGPA: **9.26**

Bachelor's in Agriculture,
Annamalai University, Chidambaram, Tamil Nadu, India.

Graduated, **May 2019**
OGPA: **8.42**

Higher Secondary Education (HSC)

A.C.S. Metric Higher Secondary School, Arni, Tamil Nadu, India.

Graduated, **March 2015**
Percentage: **91.66 %**

Secondary School Education (SSLC)

Subramania Sastriar Higher Secondary School, Arni, Tamil Nadu, India.

Graduated, **March 2013**
Percentage: **89.80 %**

THESIS

Doctor of Philosophy [Plant Molecular Biology and Biotechnology]

Evaluation of *in vitro* and *in silico* antidiabetic effects of *Costus pictus* D. Don. Ex. Lindl. and *Dodonaea viscosa* (L.) Jacq.

Master of Science [Plant Molecular Biology and Biotechnology]

Studies on *in vitro* regeneration protocol in *Costus speciosus* - a high valued insulin herb

PUBLICATIONS

Kanmani Bharathi, J., Akash, K., Rameshkumar, S., Anandan, R., & Prakash, M. (2026). Bioassay-guided discovery of antidiabetic phytochemicals from methyl jasmonate-elicited callus of *Dodonaea viscosa* (L.) Jacq.: A multitarget therapeutic approach. *Journal of the Indian Chemical Society*, 102883. <https://doi.org/10.1016/j.jics.2026.102883>

Murali, S., **Kanmani Bharathi, J.,** & Prakash, M. (2026). Standardization of tissue culture techniques for propagation of *Melochia corchorifolia* L. and analysis of phytochemical and antioxidant activity of methanol extracts." *South African Journal of Botany*, 193, 420-432. <https://doi.org/10.1016/j.sajb.2026.03.020>

Poornalakshmi, M., **Kanmani Bharathi, J.,** Sai Prathyusha Neelam, S., Gurumoorthy, P., Divya, M., Prakash., & Anandan, R. (2026). Development of an efficient in vitro system for micropropagation and regeneration of *Justicia adhatoda*, *Sida acuta*, *Pimenta dioica*, and *Premna integrifolia*. *Discover Plants*, 3(13), 1-7. <https://doi.org/10.1007/s44372-026-00477-4>

Akash, K., Parthasarathi, R., **Kanmani Bharathi, J.,** & Elango, R. (2025). Exploring the Role of Microbes in the Biodegradation of Plastic Waste: Mechanisms, Interactions, and Implications for Sustainable Waste Management-A Review. *Water, Air, & Soil Pollution*, 236(12), 1-49. <https://doi.org/10.1007/s11270-025-08343-x>

Kanmani Bharathi, J., and Prakash, M. (2024). Estimation of total flavonoids, phenols, alkaloids, tannins and in vitro antioxidant activity of *Costus pictus* D. Don. ex. Lindl. and *Dodonaea viscosa* (L.) Jacq. leaf extracts. *Chemical Papers*, 1-15. <https://doi.org/10.1007/s11696-024-03802-x>

Kanmani Bharathi, J., and Prakash, M. (2024). Phytochemical screening and quantitative analysis, FTIR and GC-MS analysis of *Costus pictus* D. Don ex Lindl. Leaf extracts-A potential therapeutic herb. *Journal of the Indian Chemical Society*, 101(11), 101362. <https://doi.org/10.1016/j.jics.2024.101362>

Kanmani Bharathi, J., Suresh, P., Prakash, M. A. S., & Muneer, S. (2024). Exploring recent progress of molecular farming for therapeutic and recombinant molecules in plant systems. *Heliyon*, 10, e37634. <https://doi.org/10.1016/j.heliyon.2024.e37634>

Priyadharshini, B., Anandan, R., Manikandan, S., **Kanmani Bharathi, J.**, and Prakash, M. (2023). Evaluation of black gram (*Vigna mungo* (L.) Hepper) genotypes for coastal saline tolerance using microsatellite markers. *Brazilian Journal of Botany*, 1-23. <https://doi.org/10.1007/s40415-023-00955-z>

Bhuvaneshwari, R., Johnny, S.I., **Kanmani Bharathi, J.**, Joshi, J.L., and Anandan, R. (2023). In vitro regeneration through organogenesis in *Costus igneus* –an important herbal insulin plant. *Research Journal of Biotechnology*, 18(3), 70-78. <https://doi.org/10.25303/1803rjbt70078>

Kanmani Bharathi, J., Anandan, R., Benjamin, L. K., Muneer, S., & Prakash, M. A. S. (2022). Recent trends and advances of RNA interference (RNAi) to improve agricultural crops and enhance their resilience to biotic and abiotic stresses. *Plant Physiology and Biochemistry*, 194, 600-618. <https://doi.org/10.1016/j.plaphy.2022.11.035>

Kanmani Bharathi, J., Anandan, R., Rameshkumar, S., Menaka, K., & Prakash, M. A. S. (2022). Development of an efficient in vitro regeneration system in *Costus speciosus*-an important herbal insulin plant. *South African Journal of Botany*, 149, 468-475. <https://doi.org/10.1016/j.sajb.2022.06.031>

Jeevitha, S., **Kanmani Bharathi, J.**, Saravanan, K. R., Prakash, M., Murugan, S., & Anandan, R. (2022). Genetic diversity and marker trait association for yellow mosaic virus disease in Green gram, *Vigna radiata* (L.) Wilczek. *Indian Journal of Experimental Biology*, 60, 481-489.

Prakash, M., Sunilkumar, B., **Kanmani Bharathi, J.**, & Viswanathan, C. (2022). Root dynamics and drought stress management in plants—An overview. *Indian Journal of Experimental Biology*, 60, 449-455.

Kanmani Bharathi, J., & Prakash, M. (2021). In vitro propagation of pharmaceutical and endangered medicinal plants-a mini review. *Journal of Current Opinion in Crop Science*, 2(4), 433-440.

BOOK CHAPTERS

Paramasivam, S., **Kanmani Bharathi, J.**, Selvamani, S., & Wang, S. (2026). In Vitro Regeneration and Genetic Transformation of Mungbean. In: Paramasivam, S., Tian, J., Pandiyan, M., Liu, C., Lixia, W. (eds) Mungbean. Springer, Singapore. https://doi.org/10.1007/978-981-95-7637-1_9

Kanmani Bharathi, J., Selvamani, S., Paramasivam, S., & Wang, S. (2026). Gene Editing in Mungbean: Opportunities and Challenges. In: Paramasivam, S., Tian, J., Pandiyan, M., Liu, C., Lixia, W. (eds) Mungbean. Springer, Singapore. https://doi.org/10.1007/978-981-95-7637-1_10

Paramasivam, S., Sasikala, K., **Kanmani Bharathi, J.**, Rajesh, S., & Pandiyan, M. (2026). Molecular Breeding Approaches for Abiotic Stress Tolerance in Mungbean (*Vigna radiata* L. Wilczek). In: Paramasivam, S., Tian, J., Pandiyan, M., Liu, C., Lixia, W. (eds) Mungbean. Springer, Singapore. https://doi.org/10.1007/978-981-95-7637-1_7

Kanmani Bharathi, J., and Prakash, M. (2024). Micropropagation of vetiver grass – A review. *Micropropagation of Medicinal Plants: Bentham books*, 1(1), 283-296. <https://doi.org/10.2174/9789815196146124010017>

CONFERENCES

International Conference on “**Recent Trends in Biodiversity Conservation**” (ICRTBC-2023) conducted by PG & Research Department of Zoology, Periyar Arts College, Cuddalore held on 12th April, 2023.

AWARDS/HONORS

Dr. A.P.J. Abdul Kalam Young Scientist Award – “9th National Conference on Addressing Environmental challenges in Agricultural Production” held at Pushkaram College of Agriculture Sciences, Tamil Nadu, India, on 29th December, 2023.

University Research Studentship – Directorate of Student Support and Progression, Annamalai University, India [2023-2024].

WORKSHOPS

AI-Assisted Drug Designing organized by Barcode Biotechnology on 30th January, 2026.

Five-Day Online National Workshop on “AI-Digital Boot Camp” organized by PMMMNMTT and TNTEU, Chennai during 30th June – 4th July, 2025.

Modelling Amorphous Solid Dispersion Release Mechanisms organized by Chemistry world and Schrodinger, 10th June, 2025.

PPV & FRA sponsored workshop on **“Training cum Awareness Programme on protection of Plant Varieties and Farmers’ Rights Act 2001”** organized by Tamil Nadu Rice Research Institute, Aduthurai, Tamil Nadu on 24th January, 2025.

Author Workshop: Publishing Open Access in Wiley Journals with Expert Tips for Authors organized by Wiley Research Training, 22nd May, 2025.

Biofertilizer and Composting Technology conducted by Shanmuga Industries Arts & Science College, Tamil Nadu, India, on 14th March 2024.

Apiculture Training Program conducted by Tamilnadu State Council for Science and Technology and Department of Entomology, Annamalai University, India, on 18 – 22nd December, 2023.

CONFERENCES/WEBINARS

International webinar on **“Costal Agriculture – An Interactive Webinar”** conducted by Faculty of Agriculture, Annamalai University on 28th August, 2020.

International webinar on **“Biodiversity Conservation and Mitigation of Impacts of Climate Change”** conducted by Faculty of Agriculture, Annamalai University on 26th August, 2020.

International webinar on **“OMICS in Crop Improvement”** conducted by Faculty of Agriculture, Annamalai University on 21th August, 2020.

Webinar on **“Nurturing Future Biotech Industry Professionals”** organized by Division of Biotechnology, Karunya Institute of Technology and Sciences, Coimbatore held from 3-8th June, 2024.

National webinar on **“Hybrid Seed Production Using Two-Line Breeding”** organized by School of Agricultural Sciences, Bharath Institute of Higher Education and Research, Chennai on 16th February, 2022.

Webinar on **“Plant Genomics and Phytotherapy”** organized by The Biomics on 12th September, 2020.

Webinar on **“Molecular Engineering: Tissue Engineering and Organ Printing and Nucleotide Excision repair in *Mycobacterium tuberculosis*”** organized by The Biomics on 05th September, 2020.

Webinar on **“Medical Health Care: BCG, a Century Old Vaccine against CoVID19: Science and Sense and Microbial Enzymes in Healthcare”** organized by The Biomics on 29th August, 2020.

Webinar on **“Demystifying COVID19: Tools of Bioinformatics for COVID19 Resarch and Immunology, Immunopathology and Immune Escapism of Viral Infection”** organized by The Biomics on 22th August, 2020.

Webinar on **“Regulatort Genomics and Epigenomics: Technologies and Applications”** organized by The Biomics on 15th August, 2020.

Webinar on **“DNA Damage Responses, Cancer Susceptibilities and Therapeutics”** conducted by Department of Biochemistry and Biotechnology, Annamalai University on 20th July, 2020.

Webinar on **“Nourish Immunity Beat Pandemic”** organized by Mather Terasa Group of Institution, Pudukkottai, Tamil Nadu on 14th July, 2020.

PEER REVIEWES

Nusantara Bioscience ISEA Journal of Biological Science – 1 (2024)

South African Journal of Botany (Elsevier) – 2 (Dec 2024; May 2025)

LANGUAGES

- English
- Tamil

PASSIONS & PURSUITS

- Tech & AI enthusiast
- Science communicator
- Script writer
- Art of cooking
- Multimedia enthusiast

REFERENCE

Prof. M. Prakash
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